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Nanobubble water promotes anaerobic digestion of high-solids cattle manure under mesophilic and thermophilic conditions

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ABSTRACT

The gradual increase in cattle farming has led to a huge production of cattle manure (CM), but the conventional treatment methods are less efficient. In this study, the treatment method of anaerobic digestion (AD) of high-solids CM by combining nanobubble water (NBW) with different gases was proposed to present a new idea for the reduction, harmlessness, and resourcefulness of CM. It was found that the performance of the digester with added NBW was better than the control. Among them, the cumulative methane yield T-Air: 227.09 mL g^{-1} VS_{added} and T-CO_2: 226.12 mL g^{-1} VS_{added} increased by 17.72 % and 17.22 %, respectively, compared with the control T: 192.90 mL g^{-1} VS_{added} under thermophilic conditions. Under mesophilic conditions, M-Air: 162.39 mL g^{-1} VS_{added} increased by 9.68 % compared with control M: 148.05 mL g^{-1} VS_{added}. Microbial communities analyzed at the genus level revealed that the relative abundance of bacteria favorable to hydrolysis and acid-producing processes, such as *Defluviitalea*, *Haloplasma*, and *Bacillus*, increased to varying degrees. Moreover, the relative abundance of archaea favorable for methanogenesis, such as *Methanoculleus*, *Methanobrevibacter*, and *Methanosarcina*, also increased to varying degrees. Therefore, the addition of NBW promoted the hydrolysis of high-solids CM, enhanced the stability of the reaction, improved the methanogenic performance, and increased the RA of favorable genera, which ultimately led to a better performance of the AD of high-solids CM.

1. Introduction

Livestock is an important contributor to the global agricultural market, and to meet the growing demand for meat, traditional decentralized family-scale livestock farms are gradually being transformed into centralized livestock farms. As a result, these farms are faced with being "flooded" with large quantities of manure that are constantly being produced (Li et al., 2021). Major cattle-producing countries include the United States, which has over 94 million head, the United Kingdom, which raises tens of millions of cows and calves annually, as well as other countries such as Brazil, with 218 million head, India, with 186 million head, Argentina, with 53 million head, and Australia, with 26 million head (Font-Palma, 2019). This situation is also adapted to China, in which in 2023 about 102 million head of cattle will be raised in the country (MAFF, 2023). It is estimated that one cow produces about 10 kg of cow manure (CM) per day (Tariq et al., 2022), so China produces about 1 billion tons of CM per day. If this CM is not managed effectively, various problems will occur, such as air pollution, water pollution, the proliferation of harmful bacteria, and the release of toxic substances (Zeng et al., 2024). Currently, the main solution is to use most of the CM as fertilizer on agricultural land to improve soil fertility and crop yield (Yan et al., 2022). However, to some extent, the long-term utilization of this method leads to the increasing accumulation of some heavy metals (e.g., cadmium and mercury) in agricultural soils (Liu et al., 2020). So improper treatment of CM may give significant health risks and environmental problems. Therefore, it is crucial to utilize appropriate methods to treat CM in a harmless, reduced, and resourceful way.

As to manure treatment, anaerobic digestion (AD) is a suitable method, as this method can simultaneously degrade organic matter, harmlessly discard wastes, and generate clean energy through a series of biological processes (Mumtaz et al., 2023; Neshat et al., 2017). Among them, high solid anaerobic digestion (HSAD) (usually referred to as digestate TS content higher than 8%) is an outstanding technology for harvesting energy from organic waste (Fan et al., 2020). This technology

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achieves a lower moisture content in the digestive system, lower wastewater production, lower digester volume, higher volumetric methane yield rate, and digestates with higher fertilizer potential (Bao et al., 2023).

The HSAD process is highly influenced by the composition of the manure, which, depending on the source, may contain a high load of recalcitrant substances or lead to the formation of toxic compounds (e.g., NH_3), which results in a slow and incomplete hydrolysis phase, a decrease in the cumulative methane yield (CMY), and, ultimately, a decrease in the energy potential of the waste (Anacleto et al., 2022). For example, CM contains a large amount of lignocellulose, and these recalcitrant parts may hinder the hydrolysis of CM in AD due to its complex structure, coupled with the fact that HSAD itself is more difficult to hydrolyze, so the use of CM as a substrate will severely limit the rate and effect of hydrolysis, resulting in a reactor that is susceptible to inhibition and longer lag periods, which will ultimately affect the yield and rate of methane (Neshat et al., 2017).

Besides, temperature is an important factor in HSAD, and HSAD exhibits different characteristics under mesophilic and thermophilic operating conditions. Mesophilic HSAD (25 °C–40 °C) has been widely used because of its good stability (Wang et al., 2023). Thermophilic HSAD (>45 °C), on the one hand, is limited in its application due to its high energy demand and poor stability. On the other hand, it is favored for its higher hydrolysis rate, fast gas production rate, and high methane production potential (Liu et al., 2022). Anaerobic co-digestion and Pretreatment (e.g., thermal pretreatments, physical pretreatments, and chemical pretreatments) (Anacleto et al., 2022; Di Capua et al., 2020; Venkateshkumar et al., 2023), are often used to improve HSAD performance. However, most of these pretreatment methods have not been applied in industries due to chemical contamination, high energy consumption, low post-waste utilization, or operational complexity (Wang et al., 2020). Simple and cost-effective methods are needed to improve CM difficult hydrolysis and digester stability during HSAD while improving methane productivity.

Nanobubble water (NBW) is a body of water that contains a large number of nanoscale particulate bubbles (Atkinson et al., 2019). It possesses properties not found in ordinary water, such as the ability to generate reactive oxygen species (e.g., O_2·-, H_2O, $^1O_2$, or ·OH), high negative ζ-potential, low buoyancy, high gas solubility, long residence time, and high surface charge. Whereas, water plays a crucial role in the anaerobic digestion process by promoting substrate hydrolysis and transferring hydrolysis products and other intermediates to places where they can be utilized by microorganisms (Kothari et al., 2014). Therefore, the concept of coupling NBW with AD is feasible. Moreover, the preparation method of NBW is simple and inexpensive, which only requires an instrument to pump gas into tap water or distilled water, which can produce 8–11 L of NBW per minute (Wang et al., 2019), and more critically, the process of NBW generation and application are both green (Chuenchart et al., 2021). So, this technology has the potential to be applied in practical engineering.

Different gases can produce different types of NBW, such as Air, CO_2, H_2, N_2, or O_2, which produce NBW that can be used in different digestion substrates. A study found that the addition of air-nanobubble water (Air-NBW) increased total alkalinity (TA) and partial alkalinity (PA) and decreased the volatile fatty acid (VFAs)/TA ratio; as a result, the stability of the AD process was maintained (Fan et al., 2021a). Nanobubbles increased water mobility and resulted in the higher mass transfer of nutrients to the microbial cells, which increased enzyme activity and decomposition rates for substrates with more CMY (Wang et al., 2019). The addition of N_2 and Air-NBW to the AD system of kitchen waste found that the activity of hydrolytic enzymes increased by 8 %–47 % in the first phase, and the activity of coenzyme F_{420} increased significantly by 34 % in the second phase (Hou et al., 2021b). The results indicated that the CMY of the digester with NBW addition of pig manure was 19 %–39 % higher than that of the control group (without NBW addition) (Fan et al., 2021b).

In summary, conventional processes for the treatment of high organic content wastes may result in disadvantages such as slow and incomplete substrate hydrolysis stages, unstable reaction systems, and lower cumulative methane production, ultimately resulting in lower energy potential of the waste. However, the unique characteristics of NBW may improve the above disadvantages. The NBW coupled HSAD process is easy to operate without adding too many chemicals and is environmentally friendly, making NBW an innovative alternative to reduce the use of chemicals and energy. However, so far, the application of NBW in HSAD has been reported less, and most of the studies on the application of NBW in HSAD were carried out under mesophilic conditions, and the performance changes of NBW on HSAD under thermophilic conditions are not clear. And this study reveals for the first time the effects of different NBW on the performance of HSAD reaction with CM as substrate under thermophilic and mesophilic conditions. So it is still unknown whether NBW can effectively improve the performance of HSAD with CM as a substrate. By exploring the mechanisms related to the changes in hydrolysis rate, stability, methane yield, and other properties of NBW-HSAD at different temperatures and different NBW, and the changes in the microbial community structure, new insights into the digestion process were improved.

2. Materials and methods

2.1. Materials

Fresh CM was obtained from Holstein cows. Before the experiment, the CM was stored in the refrigerator at 4 ± 1 °C. Waste-activated sludge was domesticated under thermophilic (55 ± 1 °C) and mesophilic (37 ± 1 °C) conditions after retrieval and then used as inoculated sludge. The physicochemical characteristics of CM and inoculum are shown in Table 1.

2.2. Preparation of NBW

NBW was produced by a nanobubble generator (JWNP, AL01S, China) through which Air, CO_2, He, and N_2 were introduced at 20 ± 2 °C. The deionized water (DW) in the vessel was recirculated into the generator recirculation system for 20 min to obtain the requisite NBW. Freshly prepared NBW is a white opaque liquid that then becomes colorless and transparent after 5–10 min of standing. This phenomenon occurs because many visible microscopic bubbles rise to the surface, rupture, and disappear. The pressure was controlled at 0.4–0.5 MPa during the production of NBW.

2.3. Experimental setup

The experimental setup was similar to that in our previous study (Liu et al., 2022). The mass of substrate in each digester was 818 g (CM: NBW = 1:1, TS was approximately 13 %), and 82 g of inoculated sludge

Table 1
Physicochemical characteristics of cow manure and inoculum.

Parameters	Cow manure	Thermophilic Inoculum	Mesophilic Inoculum
Total solids (TS %)	26.60 ± 0.09	5.52 ± 0.04	5.88 ± 0.15
Volatile solids (VS %)	10.24 ± 0.02	3.33 ± 0.06	3.59 ± 0.09
VS/TS (%)	38.49 ± 0.15	60.32 ± 0.11	61.05 ± 0.12
C (%)	18.00 ± 0.33	29.53 ± 0.12	30.86 ± 0.23
H (%)	2.38 ± 0.05	4.72 ± 0.06	4.93 ± 0.08
N (%)	0.99 ± 0.02	3.87 ± 0.03	4.50 ± 0.06
S (%)	0.28 ± 0.01	0.84 ± 0.05	0.82 ± 0.07
C/N	18.18	7.63	6.86
pH	8.16 ± 0.03	7.05 ± 0.01	7.15 ± 0.02

Note: C, N, H, and S content determination is based on the dry matter of the sample.

corresponding to the temperature was added to the digester under thermophilic and mesophilic conditions. DW (T and M) was used as the control group, and different NBW, including Air-NBW, carbon dioxide-nanobubble water (CO_2-NBW), helium-nanobubble water (He-NBW), and nitrogen-nanobubble water (N_2-NBW), were used as the experimental group. The different NBW were identified as M, M-Air, M-CO_2, M-He, and M-N_2 under mesophilic conditions and as T, T-Air, T-CO_2, T-He, and T-N_2 under thermophilic conditions. Each working condition was carried out in triplicate, and no pH adjustment was performed in this study to prevent microenvironmental variations and simplify this HSAD process, which might be applied in practice. The thermophilic digesters and the mesophilic digesters were heated in a water bath at 55 ± 1 °C and 37 ± 1 °C. Finally, each digester was shaken for 1 min at a fixed time each day.

2.4. Analytical methods

The TA, TS, pH, VS, soluble chemical oxygen demand (SCOD), and total ammonia nitrogen (TAN) of the samples were determined using standard methods (APHA, 1998). The determination of free ammonia nitrogen (FAN) in liquid samples was carried out using the formula provided by (Hansen et al., 1998). The VFAs content was determined using a gas chromatograph with a flame ionization detector (SP-7890 Plus, Ruihong, China). The volume and composition of biogas were determined by a gas flow meter (LML-2, Alpha, China) and a biogas analyzer (GA 5000, Geotach, United Kingdom), respectively. The C, H, N, and S content of the samples was determined using an organic elemental analyzer (Vario EL Cube, Elementar, Deutschland).

The HiPure Soil DNA kit (Magen, Guangzhou, China) was used for digestate deoxyribonucleic acid (DNA) extraction following the HSAD process. Microbial communities were analyzed by sequencing the highly variable V3–V4 and V4–V5 regions of the 16S rDNA gene. Bacteria were amplified with primers 341 F (5′CCTACGGGGNGGCWGCAG-3′) and 806 R (5′-GGACTACHVGGGTATCTAAT-3′), and archaea were amplified with primers Arch 519 F (5′-CAGCMGCCGCGGTAA-3′) and Arch 915 R (5′- GTGCTCCCCCGCCAATTCCT-3′) amplification. The amplified products obtained from PCR were recovered cut, and quantified using a QuantiFluorTM fluorometer. The purified amplification products were then mixed in equal amounts and connected to sequencing connectors. Subsequently, sequencing libraries were constructed, and Illumina PE250 was used for up-sequencing. Finally, the raw data obtained from microbial sequencing were uploaded to the website (https://www.ncbi.nlm.nih.gov) under the bacterial accession number PRJNA1060892, and the archaeal accession number PRJNA1060904.

2.5. Statistical analysis

The correlations were statistically analyzed using SPSS software version 27.0 for Windows. Correlation analysis was performed for the process parameters, and $P < 0.05$ was considered statistically significant. Analyses were performed based on Pearson's rank correlation, * representing $P < 0.05$, ** representing $P < 0.01$ (the smaller the P-value, the better the correlation). Orange and red show a positive correlation, light blue and dark blue show a negative correlation, and the darker the color is the greater the correlation. AD experiments and NBW characterization tests were performed using Microsoft Office Excel 2019 to calculate the mean value ± standard deviation of three parallel experiments. One-way analysis of variance (ANOVA) followed by Tukey's post hoc test was performed to evaluate the significant differences regarding performance indicators. The analysis of methane yield was fitted to the methane yield by a modified Gompertz model as our previous study (Liu et al., 2022).

3. Results and discussion

3.1. Changes in digestate characteristics

3.1.1. VFAs

VFAs are a product of hydrolysis and acidification and are a substrate used to obtain methane (Chen et al., 2022). The trend of VFAs in the thermophilic digester was similar to that of SCOD in that it increased first and then decreased (Fig. 1, Fig. 3a). VFAs in all digesters at thermophilic peaked on the fifth day. The concentration of VFAs was higher in the digesters with NBW added compared with T. The concentration of VFAs in T-Air was the highest and decreased fast. This suggests that the addition of Air-NBW may lead to more efficient hydrolysis, more complete substrate degradation, and faster conversion of VFAs to methane in the reaction, which is in line with the results of the subsequent study (Fig. 4 and Table 2). The concentration of VFAs in T-He was high but decreased slowly. The high VFAs concentration of T-CO_2, but the subsequent methane production was very good, suggesting that the VFAs concentration of T-CO_2 may have peaked five days ago and declined rapidly. The concentration of VFAs was higher in the mesophilic digester compared with the thermophilic digester, but the decline was slower. This phenomenon may be related to the faster rate of methane production under thermophilic conditions (Fig. 4 and Table 2). All of the mesophilic digesters, except M-Air, had peak VFAs concentrations on the fifth day. The lower VFAs concentration in M-Air compared with M may be due to its higher VFAs production/conversion efficiency. However, the VFAs concentration of CM under mesophilic conditions in this study was lower than that of Fan et al. (2020) using pig manure as a substrate. This is supposedly because the lignocellulose contained in CM is difficult to degrade (Haque et al., 2022).

3.1.2. pH, TA and TVFAs/TA

The pH (Fig. 2a and b) of digestate is mainly influenced by the balance between ammonia and VFAs, which depends on the balance between hydrolysis, acid production (acetic acid production), and methane yield (Bai et al., 2015). During the HSAD process, the small difference in pH between the control and experimental groups occurs because the digestion substrate ratios are identical and the CM mechanism is relatively stable, resulting in little fluctuation in pH. The pH of thermophilic digesters is higher than that of mesophilic digesters. Because the rate of hydrolysis and acidification of proteins are faster under thermophilic conditions, a large amount of organic nitrogen is converted to ammonia nitrogen, and VFAs accumulates rapidly but is degraded rapidly; thus, the pH value is generally higher than that under mesophilic conditions due to the high ammonia nitrogen under thermophilic conditions and the rapid degradation of VFAs (Li et al., 2023). The pH of the digesters (Fig. 2a) stabilized between 8.06–8.45 under thermophilic conditions and between 7.55–8.00 under mesophilic conditions (Fig. 2b). This is the pH range acceptable for metabolism by methanogens and related bacteria, and the HSAD system is relatively stable (Li et al., 2017).

TA is tied to the buffering capacity of the digester (Kawasaki et al., 2020). Under mesophilic and thermophilic conditions, so the TA of the digesters were all increased and then stabilized (Fig. 2c and d), which may be due to the fact that the CM in the digested substrate contains a large amount of alkali metals (Ca and Mg), which contributes to the alkalinity (Li et al., 2021). However, the TA contents of all samples, including those with the addition of NBW, were slightly higher than those with the addition of DW. This suggests that the addition of NBW may increase the TA content when the HSAD system is slightly inhibited, which is favorable for the recovery of the stability of the HSAD system.

TVFAs/TA is a good indicator of digester stability (Fan et al., 2020). When TVFAs/TA < 0.35, TA is sufficient to buffer TVFAs, and the AD digester can operate stably; when TVFAs/TA > 0.8, the AD digester operation is extremely unstable (Liu et al., 2022). At the beginning of the experiment, the TVFAs/TA of the thermophilic digester reached

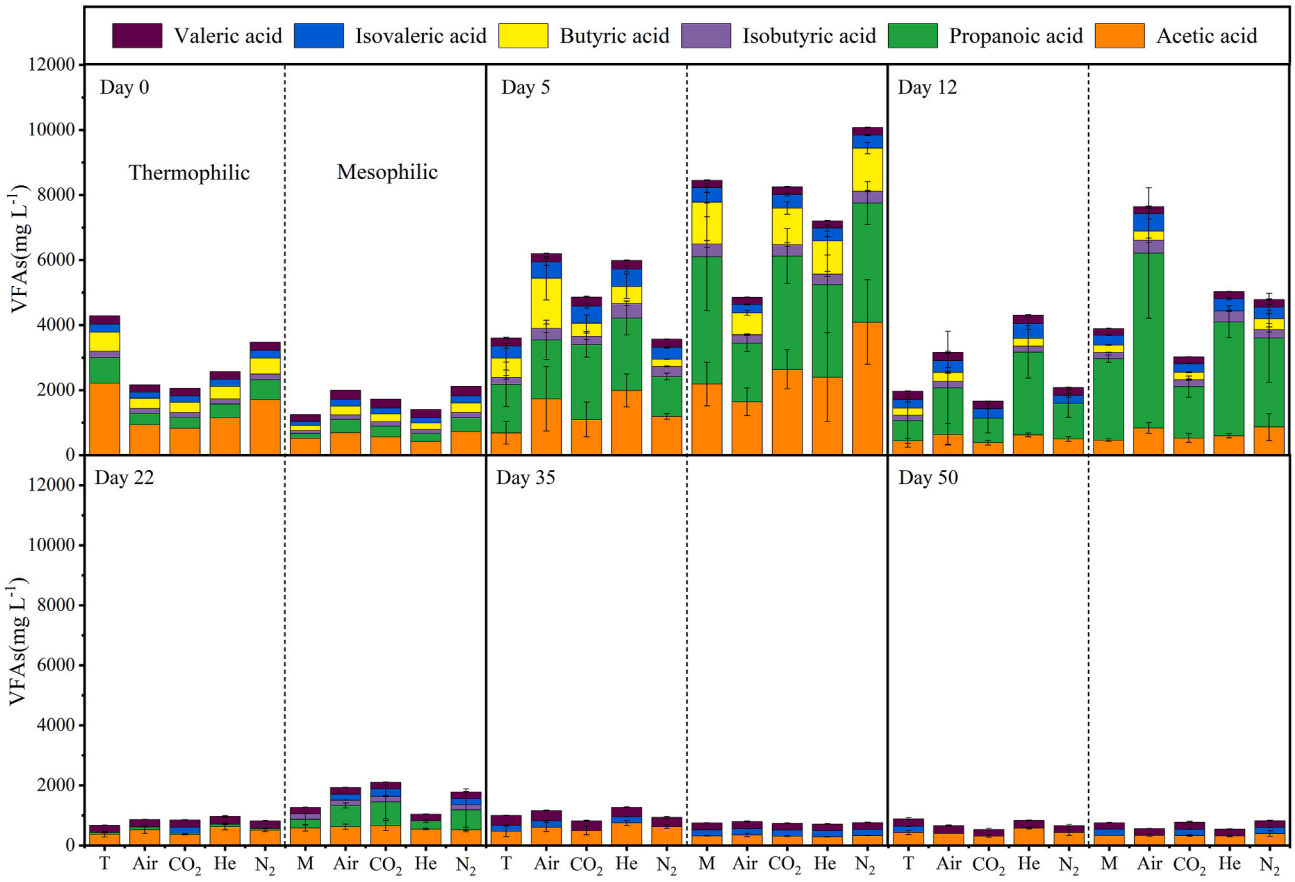


Fig. 1. Changes in VFAs during the digestion process.

Table 2
Parameters of the modified Gompertz model.

Parameters	Cumulative methane yield, CMY (mL g^{-1} VS_{added})	Maximum methane yield rate, R_m (mL g^{-1} VS_{added} d^{-1})	Lag phase, λ(d)	R^2
T	193.17	23.92	0.88	0.997
T-Air	227.56	24.33	1.10	0.997
T-CO_2	226.14	28.17	1.24	0.999
T-He	186.56	18.34	0.65	0.991
T-N_2	199.49	21.17	0.73	0.996
M	157.32	7.19	2.59	0.980
M-Air	170.42	8.23	3.05	0.989
M-CO_2	159.24	7.25	2.52	0.987
M-He	164.02	7.61	2.53	0.982
M-N_2	146.97	7.12	3.22	0.992

Note: All values are expressed as mean ± standard deviation (n = 3).

0.38–0.78 greater than the threshold value of 0.35, indicating that the digester was in an unstable state at the beginning of the HSAD reaction (Fig. 2e). Among them, the TVFAs/TA of T-CO_2 was lower, which may be due to its promotion of VFAs consumption, resulting in lower TVFAs/TA, which is similar to the change of VFAs concentration (Fig. 1). However, control T had the highest TVFAs/TA, indicating that T was the most unstable at the beginning of the HSAD reaction, suggesting laterally that the addition of NBW might make the HSAD reaction more stable. Compared with the thermophilic digester, the TVFAs/TA of the mesophilic digester fluctuated more, and the TVFAs/TA increased and then decreased in the early stage of the reaction (Fig. 2f), indicating that the hydrolysis rate of the mesophilic digester was not as fast as that of the thermophilic digester, which resulted in that the VFAs concentration in the mesophilic digester peaked only on the fifth day, resulting in the increase and then decrease of the TVFAs/TA. Among them, the TVFAs/TA of M-Air was lower, which could be attributed to the addition of Air-NBW to raise the TA and enhance the production/consumption of VFAs, similar to the previous findings (Fan et al., 2020). Finally, the TVFAs/TA of all digesters gradually decreased to below 0.35 as the HSAD reaction proceeded, which indicated that the HSAD process was stabilizing.

3.1.3. SCOD, TAN, and FAN

SCOD refers to soluble chemical oxygen demand, an important indicator of the anaerobic digestion process; the SCOD indicates the amount of soluble organic matter in the digestate (Poorasgari and Örmeci, 2022). Therefore, the hydrolytic acidification during the HSAD reaction of CM can be expressed as SCOD. The SCOD for all digesters reached their respective maximum values and then began to decline (Fig. 3a and b), because the conversion of substrate to soluble organic matter during hydrolysis increased SCOD in the first stage, and then with the consumption of organic acids by methanogen, SCOD was partially removed through biogas formation to decrease SCOD, so the differences in SCOD values were also closely related to gas production (Vu and Min, 2019). In particular, the SCOD in T-Air reached 24163.66 mg L^{-1} in the thermophilic digester on the fifth day, which was 33.4 % higher than that in T and decreased rapidly (Fig. 3a). Suggesting that the addition of Air-NBW may have promoted the hydrolysis of substrates in the HSAD hydrolysis and acidification stages. Although conducted under anaerobic conditions, the addition of Air did not stop the HSAD reaction, probably because the environment rapidly becomes anaerobic when the presence of limited trace oxygen can be rapidly consumed by aerobic or parthenogenetic anaerobic bacteria, which is similar to the results of the previous research (Xu et al., 2018). The smallest SCOD of T-CO_2 was found to be 16978.49 mg L^{-1}, but its gas production was very good, which indicated that the addition of CO_2-NBW led to faster HSAD

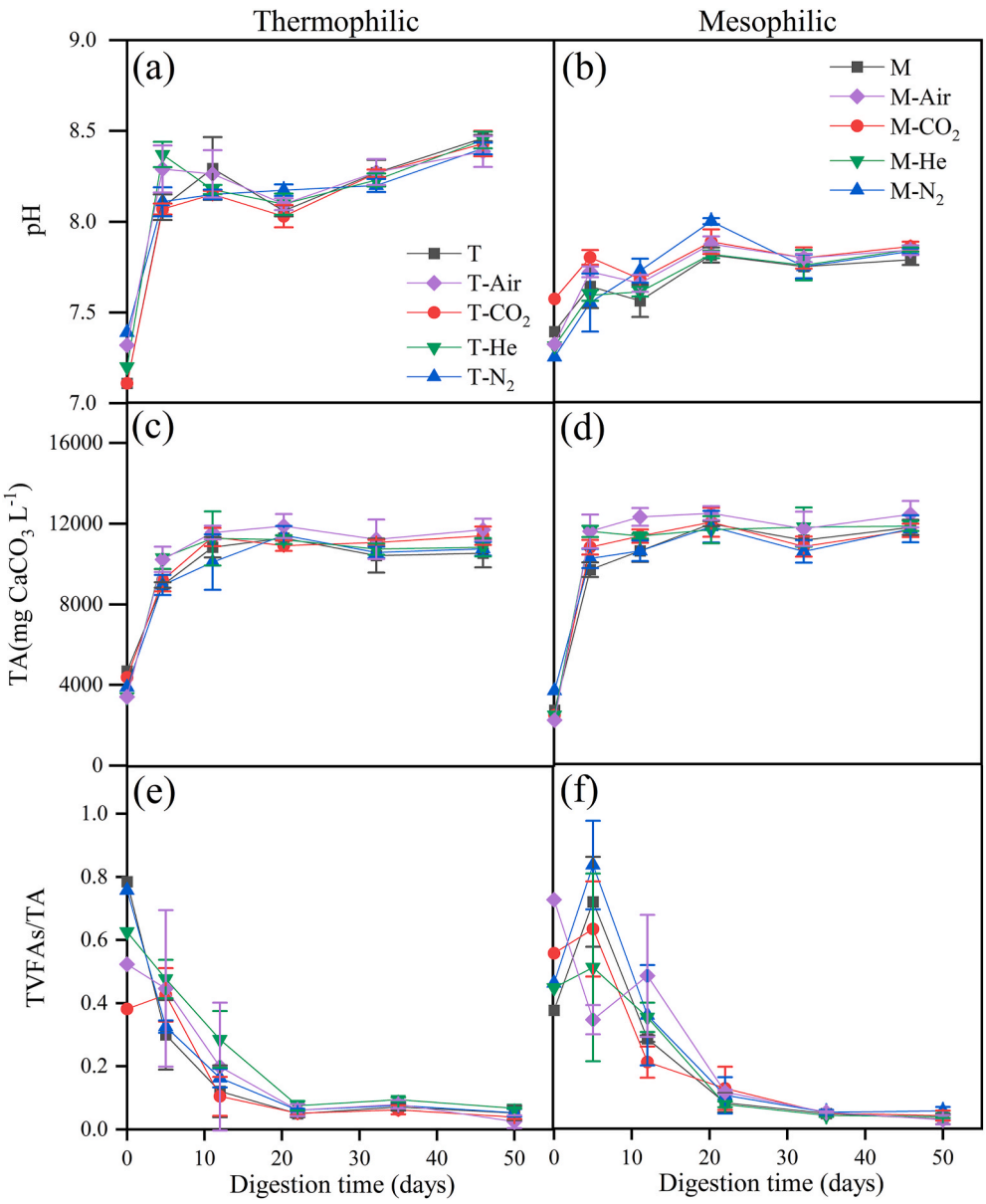


Fig. 2. Changes in pH (a and b), TA (c and d), and TVFAs/TA (e and f) during the digestion process.

hydrolysis, and the SCOD might have peaked during the first five days, which is consistent with the methane production results (Fig. 4a and Table 2). The SCOD fluctuated less in the mesophilic digester compared with the thermophilic digester, which may be due to the higher hydrolysis capacity of the thermophilic digester, resulting in the release of more nitrogen and ammonia making the HSAD reaction process potentially more unstable. The larger SCOD in the M-Air again suggests that the Air-NBW may have played a facilitator role in the hydrolysis-acidification phase (Fig. 3b).mic.

Ammonia is a byproduct of the hydrolysis of proteinaceous material (Sung and Liu, 2003). Utilizing an appropriate concentration of total ammonia nitrogen is beneficial to biological growth, as ammonia nitrogen provides a buffer and nutrients for microbial growth; however, when the ammonia concentration exceeds 1500 mg L^{-1} it will inhibit AD hydrolysis and acid production, and the activity of methanogens will be affected, which will ultimately lead to a decrease in CH_4 production (Pavez-Jara et al., 2023; Zhang et al., 2021). During HSAD, the TAN concentration in all digesters gradually increased and then stabilized (Fig. 3c and d). This may have increased the inorganic nitrogen content including ammonia nitrogen by decomposition of soluble organic nitrogen such as protein in the HSAD reaction. The concentration of TAN in each of these digesters was generally around 1500 mg L^{-1}, which indicates that most of the digesters were not significantly affected by ammonia. In addition, the ammonia nitrogen concentrations of T and M were generally lower than those of the experimental group with the addition of NBW, suggesting that the addition of NBW may have improved the hydrolysis effect, resulting in a more complete hydrolysis of the substrate leading to generally higher ammonia nitrogen concentrations. However, TAN is less toxic than FAN because FAN is able to penetrate the cell membrane of microorganisms and accumulate inside the cell, causing a proton imbalance or toxic cytoplasmic enzymes (Su et al., 2023). So excess FAN is a key toxicant for methanogenic communities during HSAD, and methanogens were largely inhibited at FAN concentrations above 250 mg L^{-1} (Sung and Liu, 2003). At the end of the HSAD reaction, the concentration of FAN in the thermophilic digester ranged from 353.04 to 419.48 mg L^{-1}, both of which were greater than the FAN threshold (Fig. 3e), and the concentration of FAN in the mesophilic digester ranged from 102.68 to 122.71 mg L^{-1}, both less than the FAN threshold (Fig. 3f). Therefore, methanogen in the thermophilic digester may have been inhibited, and methanogen in the mesophilic

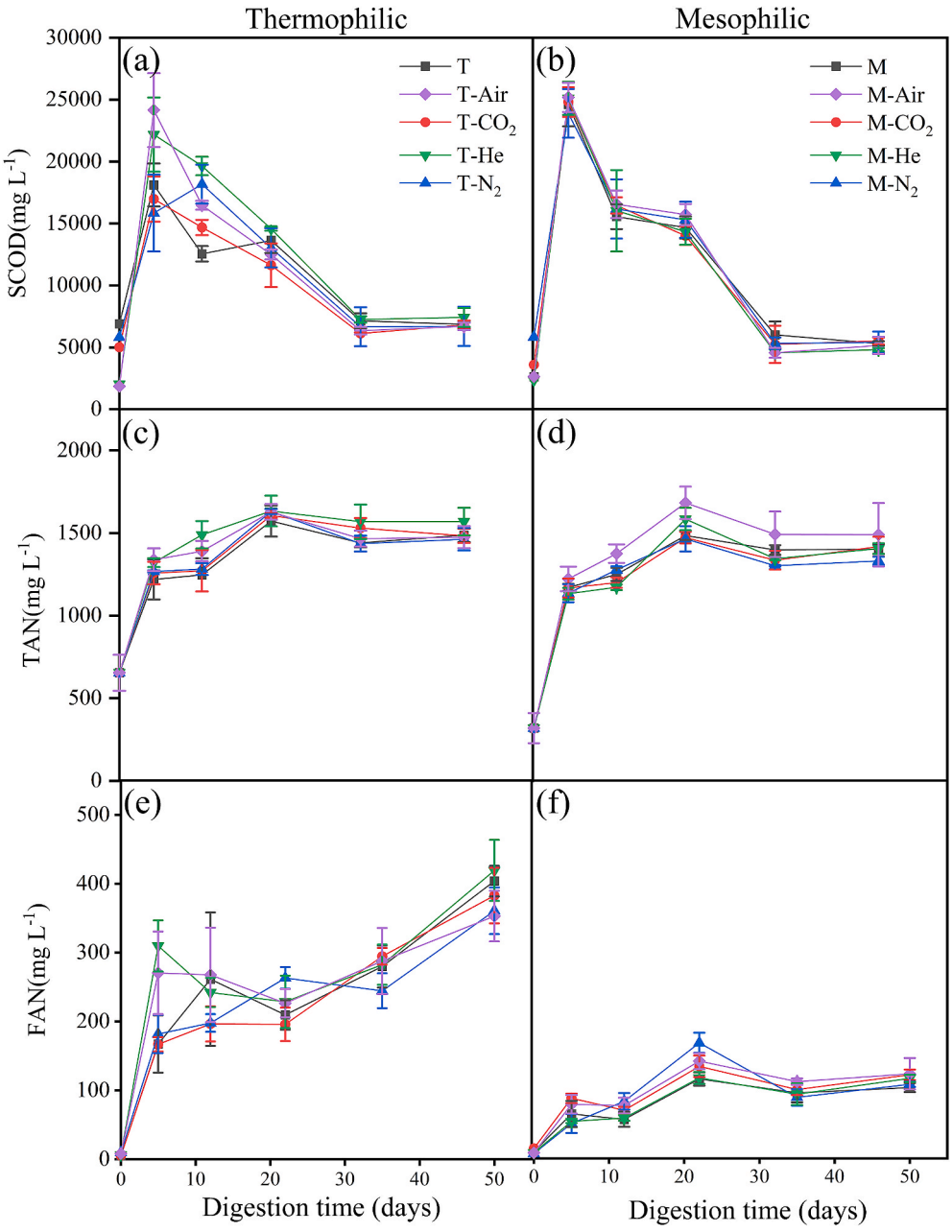


Fig. 3. Changes in SCOD (a and b), TAN (c and d), and FAN (e and f) during the digestion process.

digester was not significantly affected by ammonia concentration. The concentration of FAN in the thermophilic digester was generally greater than that of FAN in the mesophilic digester compared with the mesophilic digester because the concentration of FAN was positively correlated with pH and temperature (Wang et al., 2022).

3.2. Methane yield

Methane yield is often considered a crucial indicator for evaluating anaerobic digestion performance (Yao et al., 2017). The cumulative CH_4 yields under thermophilic conditions were as follows: T: 192.90 mL g^{-1} VS_{added}, T-Air: 227.09 mL g^{-1} VS_{added}, T-CO_2: 226.12 mL g^{-1} VS_{added}, T-He: 184.74 mL g^{-1} VS_{added} and T-N_2:198.98 mL g^{-1} VS_{added} (Fig. 4a). Among them, T-Air and T-CO_2 increased CMY by 17.72% and 17.22%, respectively, compared with T, and also produced methane at a significantly faster rate than T (Fig. 4c). Additionally, the mean CMY values of T-Air and T-CO_2 differed significantly ($P < 0.05$) from T. The cumulative CH_4 yields under mesophilic conditions as follows: M: 148.05 mL g^{-1} VS_{added}, M-Air: 162.40 mL g^{-1} VS_{added}, M-CO_2: 151.20 mL g^{-1} VS_{added}, M-He: 154.85 mL g^{-1} VS_{added}, and M-N_2: 140.37 mL g^{-1} VS_{added} (Fig. 4b). Among them, M-Air showed a 9.68% increase in CMY compared with M and also had a significantly faster rate of methane production than M (Fig. 4d). Additionally, a significant difference ($P < 0.05$) was observed in the mean value of CMY between M-Air and M. The level of methane production is also related to the stability of the system, the more stable the system is the higher the methane production is likely to be, so the addition of NBW may make the system more stable. The most favorable outcomes were observed in both thermophilic and mesophilic digesters with Air-NBW. It is noteworthy that the presence of Air leads to the death of methanogen in anaerobic digestion. However, in reality, the finite oxygen is quickly consumed by aerobic or parthenogenetic anaerobic bacteria and the digester environment rapidly becomes anaerobic (Wang et al., 2019). Additionally, some methanogens may have intrinsic tolerance to oxygen exposure and their biological

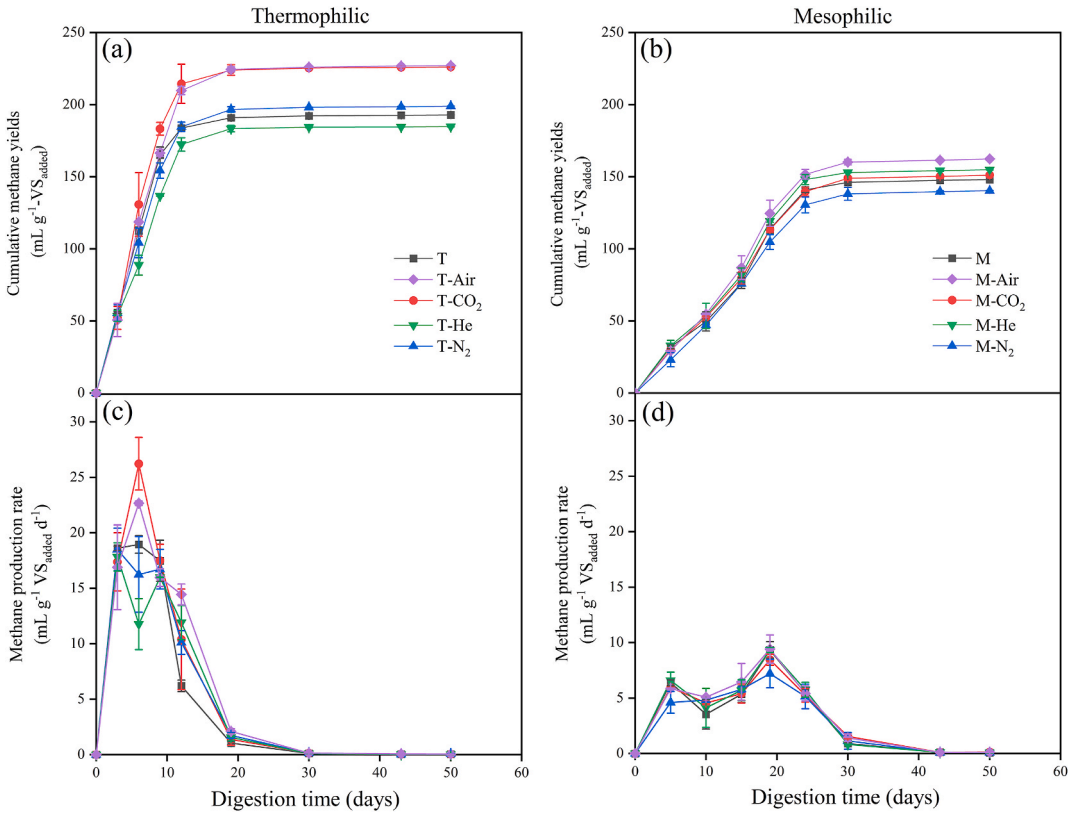


Fig. 4. Changes in CMY (a and b) and methane yield rate (c and d) during the digestion process.

activity can be quickly restored after the inhibition of oxygen exposure is removed (Botheju et al., 2010). Therefore, the addition of Air-NBW promotes anaerobic digestion significantly better than distilled water (Xu et al., 2018). The presence of limited trace oxygen may promote the hydrolysis of AD and the hydrolysis of substrates in the acidification phase, and also increase the abundance of lignocellulose-degrading microorganisms, allowing for faster and more complete degradation of lignocellulose, a key factor that impedes the improvement of hydrolysis efficiency (Chuenchart et al., 2021; Hou et al., 2021a). The additional NBW had slight but positive effects on CMY. CO_2-NBW may be readily taken up by microorganisms for anaerobic fermentation and promotion of hydrogenotrophic methanogenesis, whereas N_2-NBW and He-NBW can ensure a strictly anaerobic environment. Furthermore, the CMY and methanogenesis rates of the thermophilic digester were significantly higher than those of the mesophilic digester. This may be due to the fact that the hydrolysis stage of HSAD under thermophilic conditions is more efficient than at mesophilic, resulting in more complete substrate hydrolysis and more efficient organic matter utilization than at mesophilic.

To better analyze the dynamics of HSAD processes with different additions of NBW or DW under conditions of high-solids CM content, the improved Gompertz model was applied, and the relevant parameters were shown in Table 2. The results indicated that the experimental data fit well with the modified Gompertz model (R^2 = 0.980–0.999). The predicted maximum methane yield was consistent with the actual experimental results. Compared with the HSAD digester that contained added DW, the digesters supplemented with T-CO_2, T-Air, and M-Air showed relatively higher methane yields, and the difference was significant ($P < 0.05$), which further confirmed that the NBW addition was beneficial to methane yield. The good simulation results obtained by the kinetic model in this study showed that these models can be applied to the actual HSAD process to predict the methane yield of CM with the addition of NBW.

3.3. Characteristics of the microbial community

3.3.1. Dominant bacterial community

In the HSAD system, bacteria play an important role in the hydrolysis and acidification phases, effectively degrading macromolecules into amino acids, sugars, and alcohols and converting these small molecules into VFAs (Yu et al., 2023). However, changes in the abundance of the bacterial communities in the HSAD system were different with increasing fermentation time. In order to elucidate microbial transitions in the HSAD system, bacterial sequences at the genus level and with relative abundance (RA) greater than 1 % were selected in this study to represent the changing signs of the bacterial community.

The dominant genera under thermophilic conditions in the early stage of HSAD were *Defluviitalea*, *Haloplasma*, and *Lysinibacillus*, which accounted for 36.3 %–42.2 %, 3.4 %–6.9 %, and 0.03 %–6.8 % of the total sequences, respectively, with RA > 5.0% (Fig. 5a). Among them, *Defluviitalea* can be selectively enriched under thermophilic conditions and plays a role in the oxidation of intercalated acetates, degradation of intercalated alcohols and lactates, degradation of composite compounds, and degradation of proteinaceous substances (Jiang et al., 2020). However, the RA of *Defluviitalea* with the addition of NBW were all higher than that of T (RA = 36.3 %), and *Defluviitalea* was significantly and positively correlated with CMY and the methane yield rate ($P < 0.01$) and significantly and negatively correlated with the lag period ($P < 0.01$) (Fig. 6). This suggests that NBW may have a facilitating effect on substrate degradation, causing the digester to increase the rate of methane production, shorten the lag period, and increase CMY. As the reaction proceeded, the RA of *Defluviitalea* decreased from 36.3 %–42.2 %–0.7 % to 1.9 % (Fig. 5b), which suggests that the substrate in the digester was gradually being degraded, and the nutrient available for *Defluviitalea* to this indicates that the substrate in the digester was gradually degraded and the nutrients available for *Defluviitalea* to utilize were gradually reduced, resulting in a decrease in RA. *Haloplasma* is thermophilic and salinophilic, so it can exert physiological properties at

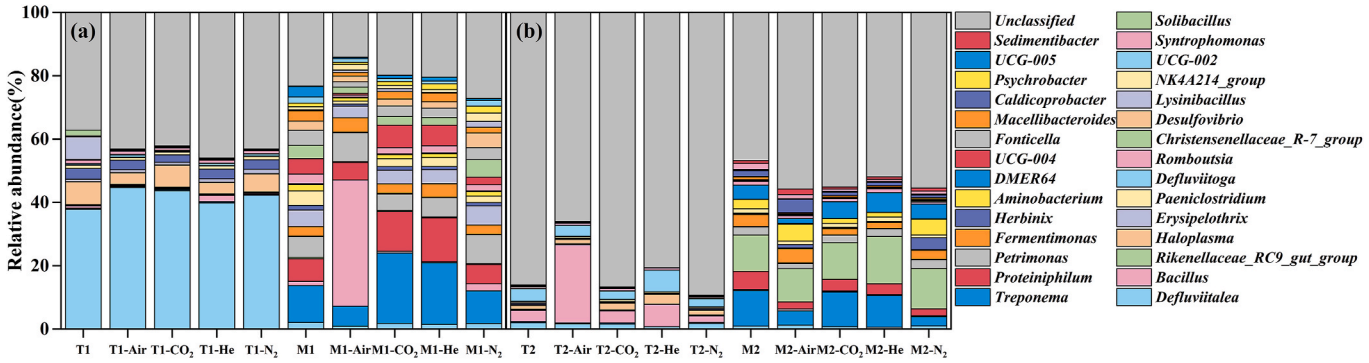


Fig. 5. Analysis of bacteria at the genus level (a: bacterial communities at the fifth-day genus level; b: bacterial communities at the fiftieth-day genus level).

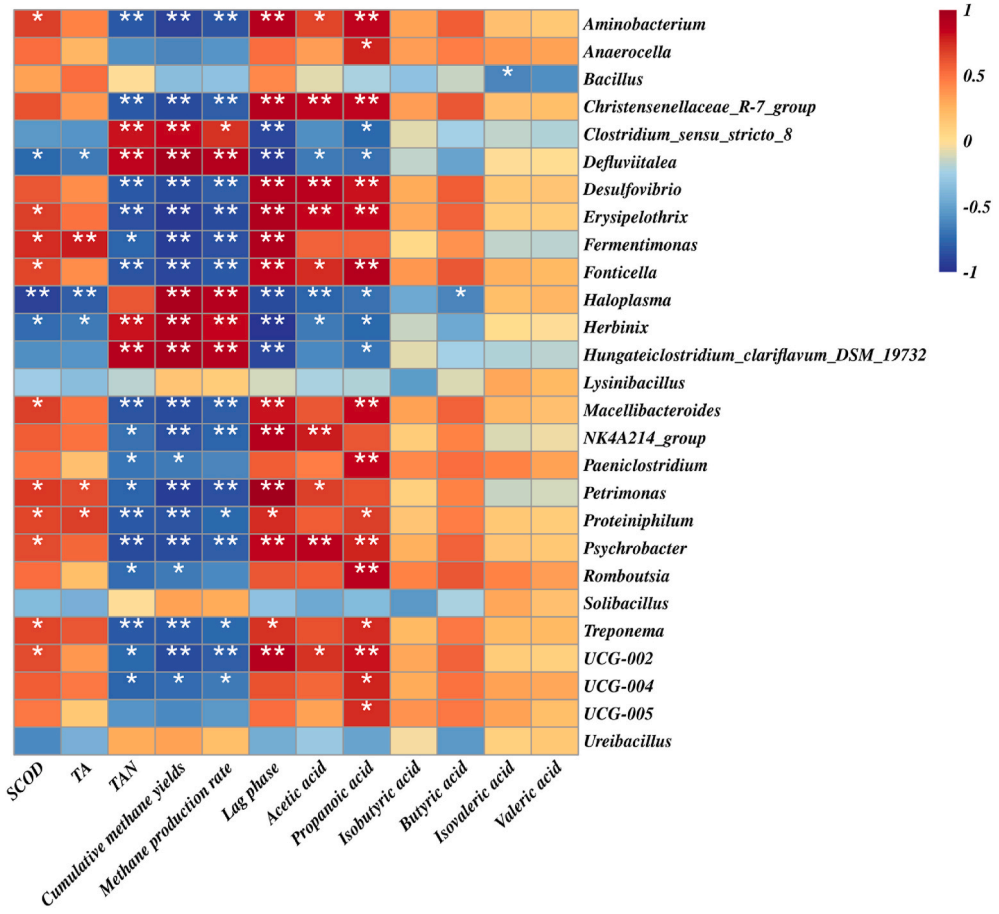


Fig. 6. Correlation analysis of bacteria (microorganisms with RA > 5 % at the bacterial genus level) with environmental factors.

high salt concentrations and thermophilic. The presence and activity of *Haloplasma* may contribute to the treatment of hypersaline wastes, promoting organic matter degradation and biogas yield (Sun et al., 2015). Additionally, *Haloplasma* showed a significant positive correlation with CMY and the methane yield rate ($P < 0.01$) and significantly and negatively correlated with the lag period ($P < 0.01$) (Fig. 6). It indicates that *Haloplasma* has a facilitating effect on methane production. The RA of *Haloplasma* in T-CO_2 is higher, so its CMY and methane production rate are higher and the lag period is shorter, which is in line with the results of the previous experiments (Fig. 4a and c and Table 2). As the reaction proceeded some inferior genera became dominant. Among them, the RA of *Bacillus* reached 2.0 %–24.5 %, and the RA of *Bacillus* in T-Air was even higher than 24.5 %, while the other digesters were only 2.0 %–7.0 %. This may be because *Bacillus* can decompose cellulose (Song et al., 2021), which accelerates the rate of hydrolysis in T-Air leading to an increase in methane production, consistent with the results of the previous experiments (Fig. 4a and c, and Table 2).

The dominant genera in the early stage of HSAD under mesophilic conditions were *Treponema*, *Bacillus*, *Proteiniphilum*, *Petrimonas*, *Erysipelothrix*, *UCG-004*, and *Christensenellaceae_R-7_group*, which accounted for the total sequences, 6.2 %–21.6 %, 0.3 %–39.2 %, 6.0%–13.4 %, 5.0 %–9.0 %, 3.6 %–6.0 %, 0.5 %–6.9 %, and 1.9 %–5.5 %, respectively, with RA > 5.0 % (Fig. 5a). Among them, *Treponema* exhibits fiber-fermenting ability, contributing to lignocellulose degradation and the ability to utilize hydrogen and carbon dioxide to convert to acetic acid, which facilitates the accumulation of acetic acid (Kawasaki et al., 2020). The high RA of *Treponema* in M-CO_2 suggests that the reactor may have better degradation of lignocellulose. Whereas members of *Bacillus* are

aerobic or parthenogenetic anaerobes that produce cellulases, hemicellulases, ligninases, amylases, proteases, and pectinases, the genus may lead to rapid oxygen depletion during the reaction and induce cellulose and lignin degradation. (Song et al., 2021). This reaffirms the feasibility of Air application in anaerobic digestion and promotes the degradation of lignocellulose, which not only accelerates the rate of hydrolysis but also improves CMY. The RA of *Bacillus* decreased as the reaction proceeded, and the RA of *Bacillus* in M-Air decreased from 39.2 % to 0.5 % (Fig. 5b). It indicates that the Air brought by the addition of Air-NBW was rapidly depleted by *Bacillus* as the reaction proceeded and the substrate was degraded to completion, leading to the decrease in the RA of *Bacillus*. Members of *Erysipelothrix*, which can be adapted to grow at high concentrations of VFAs, although dominant, do not contribute to methane yield and become dominant after system collapse (Ao et al., 2021). Whereas, the RA of *Erysipelothrix* in M and M-N_2 is higher, which may be one of the reasons for its poor methane production. In addition, the mesophilic digester had a more diverse microbial community than the thermophilic digester, both in the pre-digestion and post-digestion stages, which is similar to the results of (Chen et al., 2021).

3.3.2. Dominant archaeal community

In HSAD, archaea are the most important microorganisms in the methane yield process, and they are distinguished from bacteria by unique ribosomal RNA, the absence of peptidoglycan in the cell wall, and different membrane lipids (Harirchi et al., 2022). However, changes in the abundance of archaeal communities in anaerobic fermentation systems varied with increasing fermentation time. The variation in the archaeal community at the genus level is shown in Figures, which can be used to elucidate the microbial transition in HSAD (Fig. 7a and b).

The dominant genera in the archaeal community under thermophilic conditions in the early stage of HSAD (Fig. 7a) were *Methanobrevibacter*, *Methanosarcina*, *Methanothermobacter*, and *Methanoculleus*, which accounted for 18.8 %–48.7 %, 13.6 %–40.7 %, 10.9 %–55.8 %, and 3.3 %–14.6 % of the total sequences, respectively, with RA > 5.0 %. As these dominant genera exhibit a heat tolerance ability, they can survive under thermophilic conditions and become dominant genera. *Methanobrevibacter* and *Methanothermobacter* are hydrogenotrophic methanogens that utilize H_2 and CO_2 to produce methane (de Barros et al., 2017). *Methanosarcina* is the only methanogenic bacterium compatible with the acetate catabolic and hydrogenotrophic methanogenic pathways (Yan et al., 2022). *Methanoculleus* belongs to the group of acetic acid methanogens and can utilize acetic acid as a carbon and energy source to produce methane (Abid et al., 2021), demonstrating that hydrogen and methane yields are more effective under thermophilic conditions. Among them, T-Air and T-CO_2 had the highest methane yield under thermophilic conditions, and the RA of *Methanobrevibacter* and *Methanothermobacter* in T-Air were higher than those in the control. The RA of *Methanothermobacter* in T-CO_2 was VFAs and even higher at 55.8 %, compared with only 10.9 % in the control. This result provided a basis for its high methane yield. Furthermore, *Methanothermobacter* was positively correlated with CMY and methane production rate ($P < 0.05$) (Fig. 8), as the activity of acetic acid trophic methanogens was inhibited with increasing ammonia concentration in the later stage of HSAD. *Methanothermobacter*, on the other hand, is a hydrogenotrophic methanogen that exhibits a low sensitivity to ammonia and can perform methanogenesis at relatively high ammonia concentrations. Therefore, an increase in the RA of *Methanothermobacter* in the later stage of HSAD may lead to a concomitant increase in CMY.

The dominant genera in the archaeal community under mesophilic conditions in the early stage of HSAD (Fig. 7a) were *Methanobrevibacter*, *Methanosarcina*, and *Methanocorpusculum*. Accounting for 40.6 %–68.3 %, 11.9 %–47.4 %, and 3.2 %–37.9 % of the total sequences, respectively, with RA > 5.0 %. *Methanobrevibacter* are hydrogenotrophic methanogens that utilize H_2 and CO_2 produced in the digestion system as substrates to produce methane (Cao et al., 2022). Compared with the DW, the experimental group with the addition of He-NBW and N_2-NBW has a higher RA, which indicates that the addition of NBW exhibited a certain effect on methane yield. *Methanosarcina* is the only methanogenic bacterium compatible with the acetate catabolic and hydrogenotrophic methanogenic pathways, and the high RA of *Methanosarcina* enhances the stability of anaerobic digesters (Bao et al., 2023). However, the RA of M was only 20.3 %, and the RA of M-Air amounted to 47.4 %, which was over twice that of M. This result indicated that adding NBW led to a promoting effect on the stability and methane yield of HSAD, and M-Air is more effective. *Methanocorpusculum* belongs to the group of acetic acid methanogens that can produce methane using acetic acid as a carbon and energy source (Volmer et al., 2023). Among them, the RA of M-Air and M-N_2 were only 3.2 % and 3.3 %, respectively, suggesting that they may produce methane mainly through hydrogenotrophic methanogens.

When HSAD proceeds to a later stage (Fig. 7b), since all microorganisms require nutrients and energy to grow and reproduce, the amount of nutrients available for microorganisms decreases over time, leading to competition for resources among microorganisms (Abid et al., 2021). Faster-growing and more competitive microorganisms may occupy more resources, resulting in a lower RA of other microorganisms. The RA of *Methanobrevibacter* decreased from 18.8 %–48.7 %–11.4 %–21.1 % under thermophilic conditions. The RA of *Methanobrevibacter* decreased from 40.6 %–68.3 %–8.9 %–24.3 % under mesophilic conditions. The RA of some dominant methanogenic archaea showed a decreasing trend as the HSAD proceeded to a later stage when the VFAs and SCOD gradually decreased and the biodegradable organic matter gradually decreased.

In summary, the addition of Air-NBW influenced methanogens under mesophilic and thermophilic conditions and led to normal methane yield. Although methanogens are strictly anaerobic bacteria, the addition of air exhibits a detrimental effect on the yield of methane because microorganisms in the AD system are tolerant of oxygen (Wang et al.,

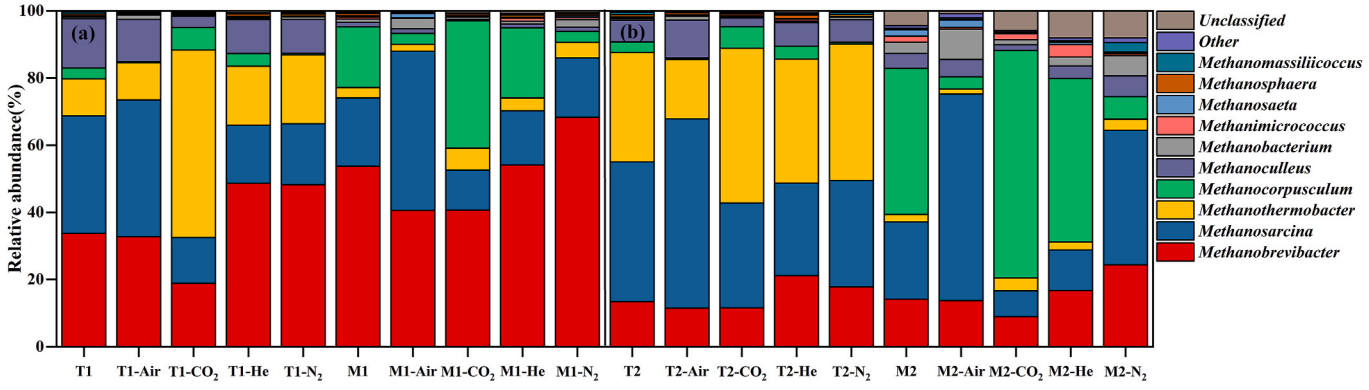


Fig. 7. Analysis of archaea at the genus level (a: archaea communities at the fifth-day genus level; b: archaea communities at the fiftieth-day genus level).

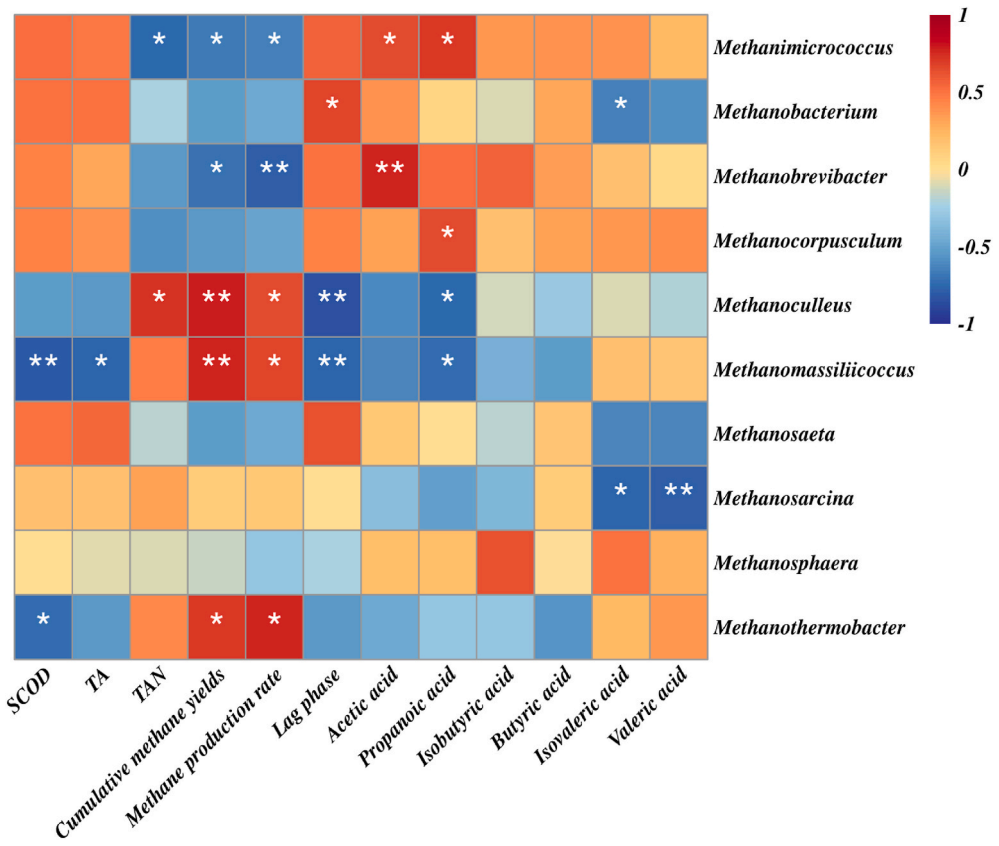


Fig. 8. Correlation analysis of archaea (microorganisms with RA > 5 % at the bacterial genus level) with environmental factors.

2019). On the one hand, facultative bacteria with rapid oxygen consumption capacity can consume dissolved oxygen, possibly depleting the added trace air during the hydrolysis phase. As a result, obligate anaerobic bacteria in anaerobic environments can survive and remain active under micro aeration conditions with no or slightly negative effects. In addition, Air-NBW primarily promotes the hydrolysis phase. On the other hand, some methanogens exhibit a certain level of oxygen tolerance, but this varies from methanogens to methanogens. *Methanosarcina*, *Methanocorpusculum*, and *Methanobrevibacter* can produce antioxidant enzymes that allow these archaea to adapt to microaerophilic conditions and exhibit high oxygen tolerance; as a result, these archaea can adapt their community structure to change oxygen levels (Cao et al., 2022).

4. Conclusions

In this study, we evaluated the changes in hydrolysis rate, digestion stability, methanogenic performance, and microbial community of HSAD by NBW with different gas species. T-Air, T-CO_2, and M-Air were the most effective in high and mesophilic digesters. Compared with T and M, the TAN concentrations of T-Air, T-CO_2, and M-Air were increased by 33.6%, 31.8%, and 34.7%, respectively, suggesting that T-Air, T-CO_2, and M-Air may promote hydrolysis and hydrolyze the substrate more thoroughly. Cumulative methane production of T-Air: 227.09 mL g^{-1} VS_{added}, T-CO_2: 226.12 mL g^{-1} VS_{added} increased by 17.72 % and 17.22 % respectively compared to control T: 192.90 mL g^{-1} VS_{added}, and M-Air: 162.39 mL g^{-1} VS_{added} increased by 9.68 % compared to control M: 148.05 mL g^{-1} VS_{added}. This indicates that NBW improved the methane production performance, and the increase in methane production also confirms the good stability of the system from the side. Also, the TVFAs/TA and FAN concentrations were smaller compared to T and M, indicating the improved stability of the system. Microbial community analysis revealed different increases in the RA of hydrolysis- and acid-producing bacteria such as *Defluviitalea*, *Haloplasma*, and *Bacillus* at the genus level, as well as the RA of methanogens such as *Methanoculleus*, *Methanobrevibacter*, and *Methanosarcina* had different degrees of increase. NBW of the remaining gases had a smaller effect but still had a positive effect compared with the control. The results showed that the addition of NBW promoted the hydrolysis of high-solids CM, enhanced the stability of the reaction, improved the methanogenic performance, and increased the RA of favorable genera. However, the mechanisms of the effects of NBW on HSAD were different for different gases, so the NBW-based HSAD system deserves further investigation.

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CRediT authorship contribution statement

Qingfang Zhang: Validation, Supervision, Methodology, Investigation, Funding acquisition. **Ruoyu Wu:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation. **Ziying Xu:** Writing – original draft, Resources, Investigation. **Yutong Feng:** Resources, Investigation. **Cheng Peng:** Resources, Investigation. **Hailong Shi:** Resources, Investigation. **Yuqian Zhang:** Resources, Investigation. **Linhai Yang:** Resources, Investigation. **Dan Luo:** Resources, Investigation. **Peng Dong:** Resources, Investigation. **Wenqi Gao:** Resources, Investigation. **Tianfeng Wang:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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